RPMGLOBAL

AMT - Modelling an Equipment

Training Manual

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RPM SOFTWARE – ASSET MANAGEMENT

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Preface

The training workshops are developed to supplement instructor-led training. The guide is constructed in a way that teaches students the key functionality of the application through practical, hands-on workshops. Each workshop details tasks that the student is required to complete. The workshops build on each other to help ensure a comprehensive understanding of the application is achieved at the completion of the course. Space is provided at the end of each workshop for learning checks and student notes.

Symbol Library



Note

A note for participants about the task.



Important Reminder

An important reminder for the participant about the task.



Learning Check

Check your understanding on the workshop and ask any related questions.

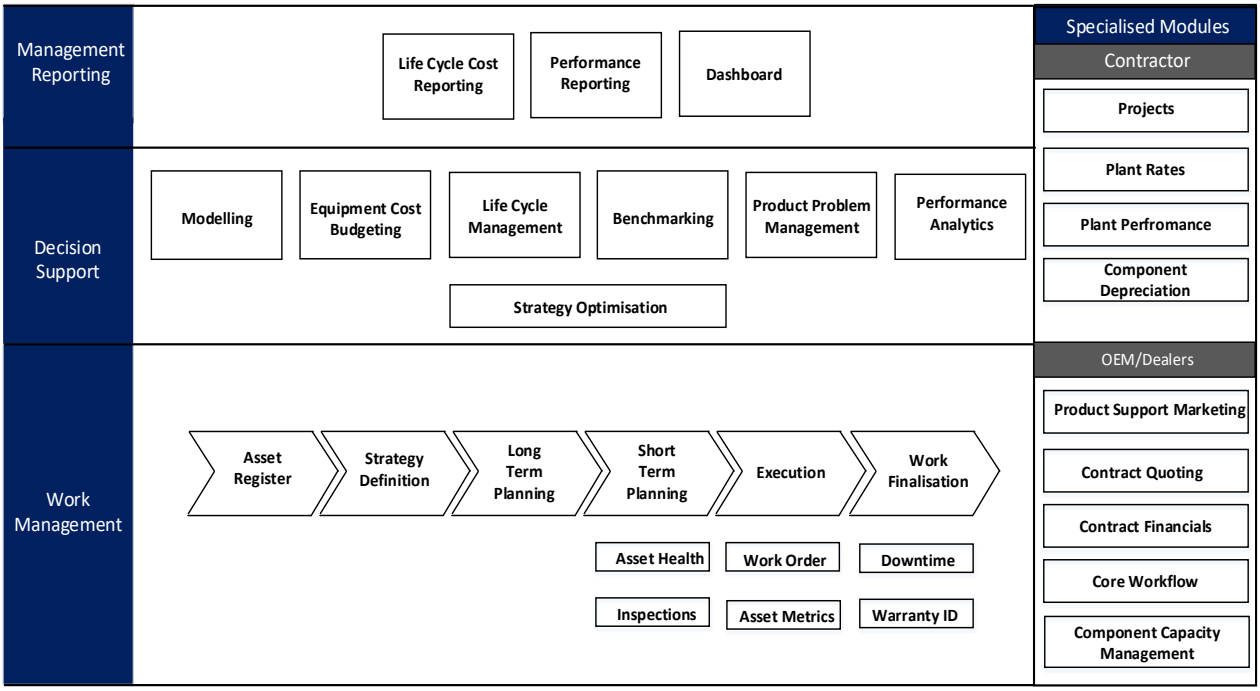
Course Outline

Pre-requisites	Getting Around AMT, AMT Key Concepts.
Target Audience	Asset Group, Contract Estimators, Project Estimators
Nominal Duration	
Course Objectives	
Learning Outcomes	At the end of the course, participants will have learned to: • Basic setup: o Review modelling-related system tables. o Create modelling hierarchies (branch/site/fleet). o Manually register a modelling equipment. o Set the term for a modelling equipment. o Copy/import/create strategy tasks. o Add a new Strategy Task (if required). o Create a Standard Job (if required). o Create Alternate Projections (if required). o Adjust Strategy Tasks. • Usage profiles: o Define Usage Assumptions. o Set Availability Forecast. o Link assets to usage profiles. • Rollover Real Assets to Modelling Assets.

	• Change Modelling Equipment o Change Cost Centre o Change Cost Centre for Individual Tasks o Change Cost Centre for All Tasks • Validate Life Cycle Costs • Link Component Structures • Create an Overhead modelling Projection
Learning Outcomes (cont.)	• Set up a modelling overhead projection. • Analyse modelling equipment.
Assessment Strategy	The completion of workshop tasks.
AMT version	8.26

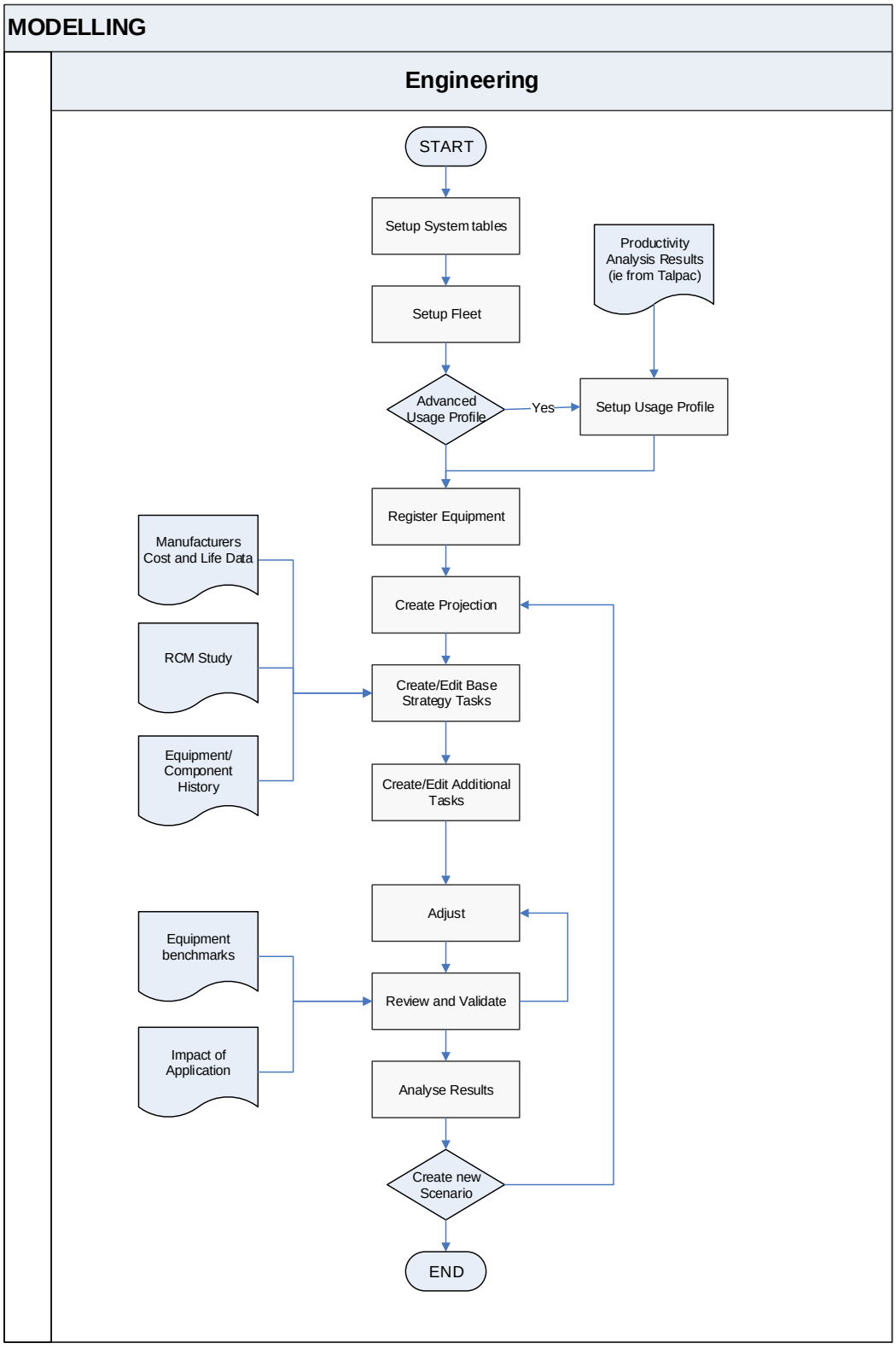
Introduction to AMT

AMT is a comprehensive asset management system that consists of integrated modules as depicted below. The functionality covered by this training manual is identified by the **Modelling** section in the diagram.



Processes

Process Diagram



Process Inputs

The following inputs are required for this process:

- Details of the cost, resources and intervals of maintenance tasks necessary to keep the equipment performing to its designed performance level. It is important to understand the context of the estimate in relation to application, environment, maintenance personnel skill level, and pricing regime.
- Historical records detailing the actual achieved maintenance task cost, resources and intervals. It is important to understand the context of the historical data to correctly apply it (that is, what was the application, environment, maintenance personnel skill level, pricing regime?).
- Analysis of the risks inherent at a particular project where the equipment is to be used. The risk analysis will consider the proposed project's application, environment, organisation, resources, market and equipment design.

The Create Modelling Assets process supports two processes primarily, the result of which may be included in a site's AMT budget or forecast:

- Modelling the potential movement of one asset to a different site.
- Modelling potential new assets.

Process Outputs

The following outputs can be expected from this process; these outputs are analyses of:

- Costs
- Labour requirements
- Parts, components and consumable requirements
- Productivity or availability

Training Workshops

1. Set Up System Tables

Procedure – Set Up System Tables

To create a modelling asset, basic system table data such as model and manufacturer details is required. Such entries may need to be created in AMT's system tables to support the modelling.

In this step, the existing system tables are updated. This update can involve changes to equipment-related maintenance and usages. The list of system tables below should be reviewed in case additional options are needed.

Typically, the AMT Administrator has access to make these changes, so a formal request to that person may be required.

Table Type	Table Name	Description
Equipment	Equipment Class	High level grouping of equipment (such as Mobile, Fixed Plant).
Equipment	Equipment Group	Subsection of equipment class (such as Off Highway Truck).
Equipment	Manufacturer	Company that builds the asset (such as Boeing).
Equipment	Model	Model of equipment (such as A380, RH400, 793C).
Usages & Measurements	Units of Measure	Life cycle measurement unit (such as engine hours, fuel).
Maintenance	Cost Type	Grouping of different categories of cost (such as repair and maintenance, finance, operator).
Maintenance	Component Code Hierarchy	Specific cost items within each cost type (such as engine, transmission, undercarriage, fuel, tyres, insurance, registration).
Maintenance	Modifier Code	Position code (such as front, rear, top, bottom, front left, front right).
Maintenance	Job Code	Describes the action performed (such as remove and install, recondition, service, weld).

2. Set Up Equipment Hierarchy

Overview

For modelling equipment, branch, site and fleet may not refer to a geographic location in the typical sense. Rather, it could be a group of equipment types for a specific tender. For example:

- **Branch** = Feasibility Studies
- **Site** = Gold Mine Evaluation – June 2020

One or more equipment can belong to a fleet. For modelling, fleets can be used to refer to manufacturers, a specific scenario or a tender. Fleets contain the availability assumptions – calculation method, tasks to include in the assumptions and calculation assumptions. Fleets also contain the standard job pricing assumptions. These assumptions can increase or decrease standard job pricing by a factor.

For example:

- **Branch** = Modelling
- **Site** = BHP Olympic Dam Tender
- **Fleet** = 150T haul trucks

Procedure – Set Up Equipment Hierarchy

Perform the following steps to validate, edit or create the modelling equipment hierarchy:

1. To validate that the branch you want exists, in the **Search** field, search for **Branch Administration**.
2. Double-click **Modelling** > **Branch Administration**.



*Alternatively, from the **My Functions** hierarchy select **Modelling** > **Administration** > **Equipment Hierarchy** > **Branch Administration**.*

3. Click **Apply Filter** on the **Branch Administration** dialog.
4. Check that the branch exists for the modelling that you want to create. If so, click **Close**.
5. If the branch does not exist, click **Add**.
6. Enter the **Branch** name in the **Add Branch** dialog and click **Save**.



If the branch created in AMT does not exist in the ERP, the labour rates need to be maintained manually in AMT. However, if standard jobs are not used to support modelling, this consideration does not apply.

7. To validate if the site you want exists, in the **Search** field, search for **Site Administration**.

   Under **Modelling**, double-click **Site Administration**.



*Alternatively, from the **My Functions** hierarchy select **Modelling** > **Administration** > **Equipment Hierarchy** > **Site Administration**.*

8. Click **Apply Filter** on the **Site Administration** dialog to view all the sites under the modelling category.

9. Check that the site exists for the modelling that you want to create. If so, click **Close**.

10. If the site does not exist, click **Add**.

11. On the **Add Site** dialog, perform the following actions:

    - Enter the **Site** name and the associated **Branch** name as required.
    - Set the **Currency** (**AUD** in the example).
    - Click **Save**.

12. To validate that the fleet you want exists, search for **Fleet Administration**. Double-click **Modelling** > **Fleet Administration**.



*Alternatively, from the **My Functions** hierarchy select **Modelling** > **Administration** > **Equipment Hierarchy** > **Fleet Administration**.*

13. Click **Apply Filter** on the **Fleet Administration** dialog to view all the fleets in AMT and validate if the fleet you want exists.

14. Check that the fleet exists for the modelling you want to create. If so, click **Close**.

15. If the fleet is missing, you need to create it. To create a Fleet, click **Add**.

16. On the **Add Fleet** dialog:

    - Select **Fleet**.
    - Select **Site**.
    - Select **Pricing Group**.
    - Set the **Service Interval**: example 250 Hrs (it is mandatory in AMT).
    - Set the **Tolerance:** example 10% (it is mandatory in AMT).
    - Click **Save**.



If standard jobs are used in the modelling, the parts pricing applied will be based on the modelling fleet's pricing group. However, if standard jobs are not used in modelling, this consideration does not apply.

*The **Service Accuracy Details** section has zero impact on LCC. However, the fields in this section are mandatory to fill in AMT. These fields are only relevant for the work management functionality of AMT.*

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

3. Create Modelling Equipment

Overview

Every modelling equipment is linked to a fleet, which belongs to a site, which belongs to a branch. Before creating modelling equipment, ensure that the relevant fleet, site, or branch to which the modelling equipment belongs exists in AMT. Remember that labour rates are inherited from the branch to which the fleet is associated, the parts price group is determined per fleet, and the currency per site.

To confirm the hierarchy, click **Modelling Equipment** ( ). This will display:

- the hierarchy (if it has been created and a modelling equipment already exists in it).
- the list of modelling equipment that already exist in the hierarchy.

If a modelling equipment does not exist in the hierarchy, there will be nothing to show in the area outlined as in the image.

After confirming the hierarchy, you can create modelling equipment. There are two ways to create modelling equipment in AMT:

- Create from an existing equipment (centrelines or real equipment).
- Create from blank.

Procedure - Create Modelling Equipment from Centreline / Real Equipment

Perform the following steps to create a modelling asset by copying a centreline or real equipment:

1. Search **Copy and Create Equipment - Modelling** using the **Search** field. Double-click **Copy and Create Equipment - Modelling**.

*Alternatively, from the **My Functions** hierarchy select **Modelling > Administration** > **Equipment Hierarchy** > **Copy and Create Equipment - Modelling**.*

*You can also select **Setup & Admin** > **Equipment** > **Copy and Create Equipment.***

*The template information to build a modelling asset can be drawn from centrelines using the **Copy and Create Equipment - Modelling** function.*

2. On the **Copy and Create Equipment** dialog, perform the following actions:
    - Select source **Equipment**. The source must be real or centreline equipment.
    - Select the **Projection Type**.
        - If a **Centreline** equipment is selected, it will have a centreline projection.
        - If a **Real** equipment is selected, it will have the current projection.
    - Enter the **Destination** for the new modelling equipment in the **Fleet** field.
    - Ensure that the **Equipment Type** is set as **Modelling**.
    - In the **Generic Details** section:
        - The **Annual Utilisation** field refers to the usage of equipment based on hours that is determined from the centreline. Enter an appropriate value.
        - The **End Usage** field refers to how long the modelling equipment will last. Enter an appropriate value.
        - The **Pricing Date** field refers to the date until when the pricing is valid. Select the desired date.
        - Select the **Multi-Projection** check box. This enables having different versions of the modelling equipment.
    - In the **Equipment Details** section, click **Add** (**+**) to add an equipment.
3. AMT adds the equipment displayed in the list area of the **Equipment Details** section.
    - Enter a unique name for the modelling equipment in the **Equipment Name** column.
    - Select dates for each modelling equipment in the **Start Date** column.
4. Click **Process**.



*You can add multiple modelling equipment to this list by clicking **Add** (**+**).*

*To remove entries from the Equipment Details section, click **Delete** (**–**).*

*If different **Start Dates** are not selected, AMT forecasts the equipment to arrive on one date and everything will start on that date. The dates can be changed later.*

5. The **Copy Projection** dialog displays. Select the **Leave the pricing as is** option and click **OK** to maintain the baseline of the pricing data.
6. The **Copy and Create Equipment** dialog displays. Click **OK**.

Note:

- The strategy tasks, and any linked standard jobs, are copied into the modelling equipment.
- When a strategy task, linked to a standard job, comes into a modelling equipment from another equipment (such as a centreline), the cost values on the modelling equipment's strategy task are a snapshot of the cost values of the source equipment's strategy task.
    - The cost values of modelling equipment are not automatically updated to reflect the modelling equipment's labour rates (sourced from the branch) or

parts pricing group (sourced from the fleet) unless the modelling equipment is under the same branch-fleet as the source equipment.
    - Moreover, the costs values are not automatically updated even if the linked standard job is updated. This is a design principle of modelling equipment in AMT.
- If you do want to see the strategy task updated with the latest standard job cost, follow the steps below:

    - Search **Refresh Standard Job Pricing** using the **Search** field and double-click the **Refresh Standard Job Pricing**.

      Alternatively, from the **My Functions** hierarchy, select **Modelling > Administration > Projections > Refresh Standard Job Pricing**.
    - The **Refresh Standard Job Pricing** dialog displays. Select the appropriate options from the **Fleet**, **Equipment** and **Projection**.
    - Populate the **Task Type** and **Strategy Tasks** fields.
    - Click **Refresh Pricing**.
    - Close the dialog.

Procedure - Create a blank Modelling Equipment

Blank modelling equipment can be created using the **Create Blank Equipment - Modelling** function.

Perform the following steps to create a blank modelling asset:

1. Search **Create Blank Equipment - Modelling** using the **Search** field.
   Double-click **Create Blank Equipment - Modelling**.



*Alternatively, from the **My Functions** hierarchy, select **Modelling > Administration** > **Equipment Hierarchy** > **Create Blank Equipment - Modelling**.*

2. On the **Add Equipment** dialog, perform the following actions:
   - **Equipment Name**: Enter a name for the equipment. We recommend using a naming convention to provide information about the equipment, such as Manufacturer-Model- and any other details. For example, MOD-KOMASTU-HT-830E-1AC.
   - Select a **Fleet**.
   - Select a **Manufacturer**.
   - Select the **Model**.
   - Choose **Utilisation: Simple** or **Advanced**.
     - **Simple**: Only one unit of measure for the asset, for example, dates.
     - **Advanced**: More than one unit of measure.
   - Select the **Multi-Projection** check box.
3. To set utilisation assumptions, click the **Term** tab and perform the following actions:
   - **Primary UOM:** From the Result List dialog, double-click to select the **Primary UOM**.
   - **Annual Utilisation:** Enter the annual utilisation.
   - **End Usage:** Enter the end usage.
4. Click the **Classifications** tab. In the **Default Cost Bearer** field, select the default cost bearer from the **Result List** dialog.
5. To save the changes:
   - Click the **Term** tab.
   - Click **Refresh**.
   - Click **Save**.

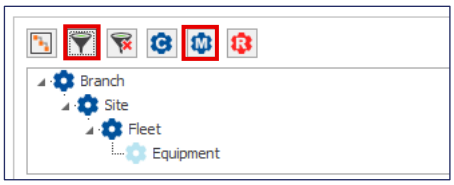
   The modelling equipment is created.

Procedure – Define life-cycle term and confirm approximate start date and usage

In this step, you define the modelling equipment's start date, usage and end date. AMT requires these details to generate an accurate life cycle costing.

Perform the following steps to define the life cycle term and confirm approximate start date and usage:

1. List the modelling equipment that you can access:
    - Click **Modelling Equipment**.
    - Click **Apply Filter**.



2. On the **Equipment Selector** dialog, populate the **Branch** and **Site** fields to help you find the modelling equipment you want.
3. Click **OK**.
4. Expand **Branch**, **Site** and **Fleet** in the **Equipment Navigator**. Double-click the required modelling equipment.
5. On the **Edit Equipment** dialog, under the **Base Details** tab, you can see the details of the created equipment, including the information inherited from the centreline or the real equipment.

    If the modelling equipment was created using the **Create Blank Equipment - Modelling** function, the following information needs to be populated:
    - In the **System Configuration** section, set the **Utilisation** section to **Simple Utilisation** (only valid for a blank modelling equipment).
    - In the **Equipment Projection List** section, you can see the projection details of the equipment.



*If the modelling equipment was created from a real equipment, the utilisation will be set to **Advanced**, whether the real equipment is linked to a usage profile or not. You will not be able to change the utilisation. If a blank modelling equipment was created, you can select simple or advanced utilisation.*

6. Click the **Term** tab. Set relevant details for:
    - **Annual Utilisation**
    - **Start Date / Start Usage**
    - **End Usage**

You can either enter a usage profile or select an existing usage profile, if suitable.

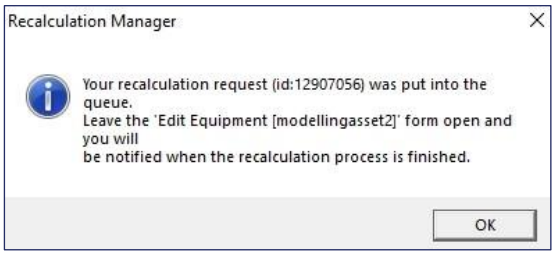
7. On the **Classifications** tab, select a valid **Default Cost Centre** from the list field.

8. On the **Tasks** tab, click **Apply Filter**. The Recalculation Notification dialog displays. Click **Yes**.

*If you click **No**, the list of tasks copied from the centreline or the real equipment displays.*

*If the modelling equipment was created using the **Create Blank Equipment - Modelling** function, the tasks should be added individually.*

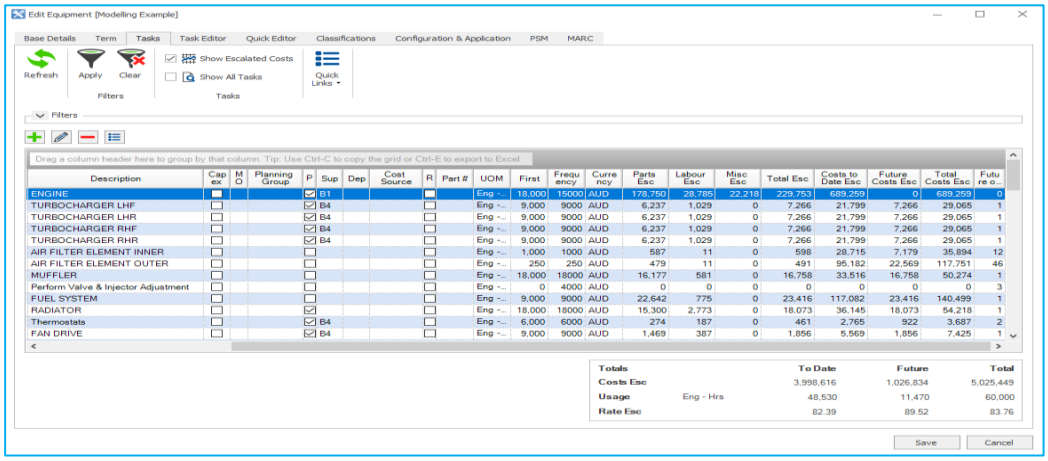
9. The **Recalculation Manager** dialog displays. Click **OK**.



10. The **Edit Equipment** dialog displays. Click **OK**.

11. AMT displays the list of strategy tasks. The future cost is calculated by multiplying the cost of the task and the number of future occurrences, that is, **Total Esc** x **Future Occs** = **Future Costs Esc**.

    - The summary of the calculation displays at the bottom right side of the dialog.



Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

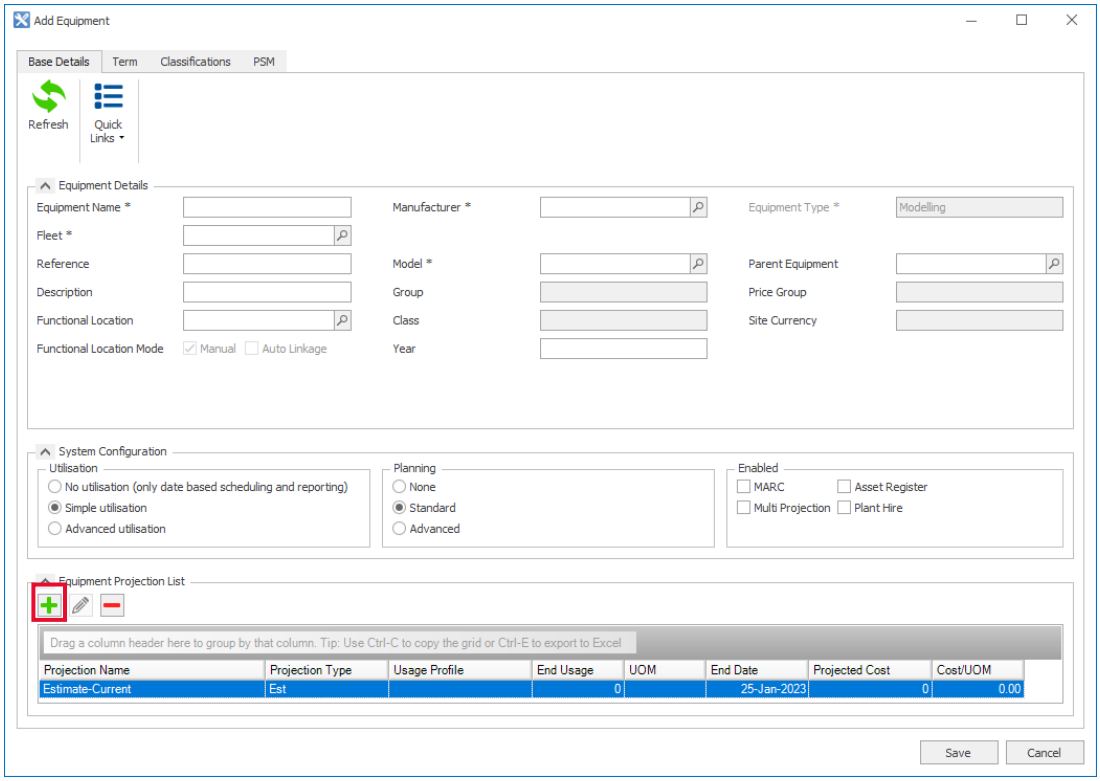
4. Create Strategy Tasks

Procedure – Copy a Projection

Now that the modelling equipment is registered, its costs have to be built up. The costs are recorded as strategy tasks in the equipment's projection.

If a relevant centreline or real equipment registered in AMT has a suitable projection on which to base the modelling equipment, it can be copied to the modelling equipment's projection (if not already done using the **Copy and Create Equipment – Modelling** function). Otherwise, the projection can be imported from an external system or load sheet, or the strategy tasks for the centreline's projection can be created as new.

1. Click **Add**.



2. Select the **Strategy Task Type**, and click **OK**.
3. Set **Projection Type** to be **Estimate – Current.**
4. Select **Use Existing**.
5. Select an **Equipment** and a **Projection**.
6. Click **Process** to copy the selected equipment's projection into the modelling equipment's **Estimate-Current** projection.



This process deletes the modelling equipment's existing projection and replaces it with the selected equipment's projection. Cost sources from the source projection will be maintained.

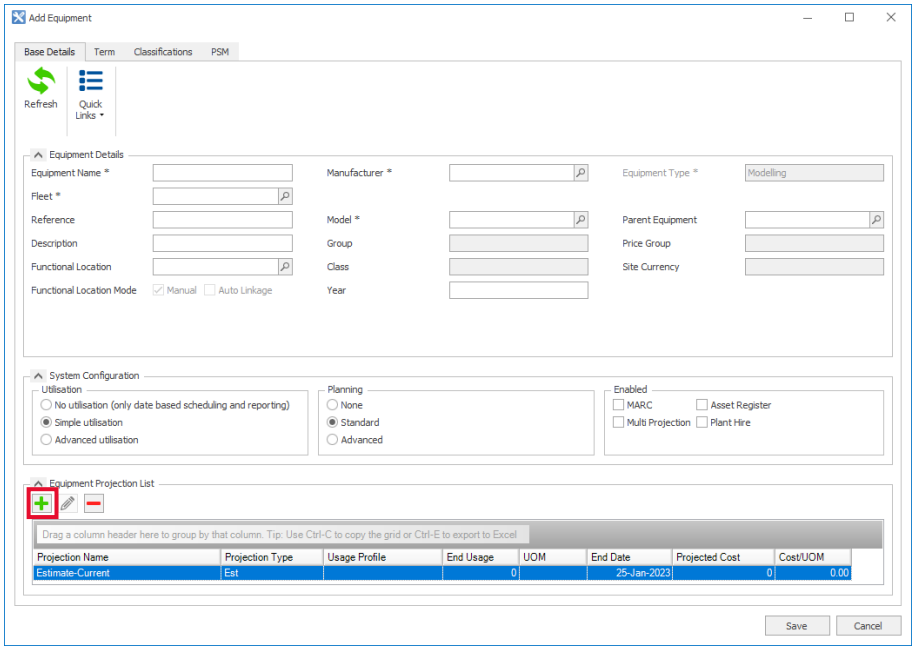


Depreciable asset structures are not copied.

Procedure – Importing Strategy Tasks

Perform the following steps to import strategy tasks:

1. Click **Add**.



2. Set **Projection Type** to be **Estimate - Current**
3. Select **Import Projection**.
4. Click the **Ellipsis**, browse to and select the file to import.
5. Select the **Equipment** and its **Projection** from the file.
6. Click **Process** to import the selected file into the modelling equipment's Estimate-Current projection.

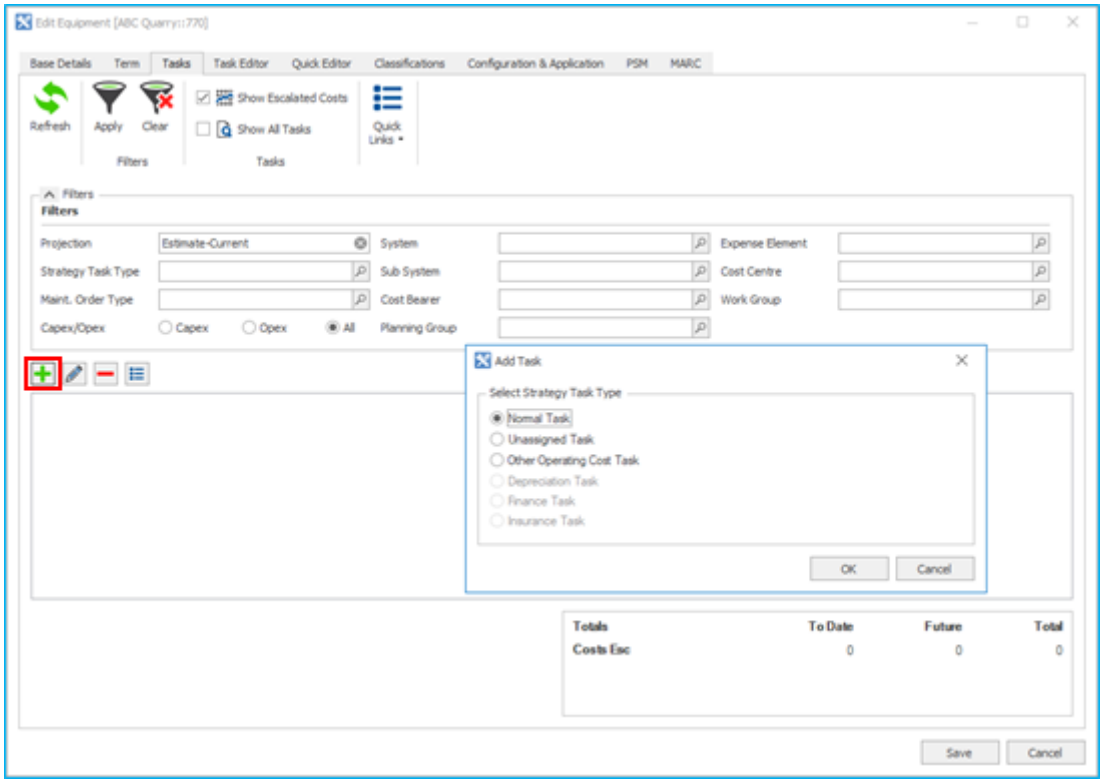
The import file must be a CSV file of a specific structure. Contact your AMT system administrator for more information.

*If errors show when clicking the **Equipment** filter, it means that the file import file is not formatted correctly, resulting in AMT not being able to locate specific equipment in the file.*

Procedure – Creating New Strategy Tasks

Perform the following steps to create new strategy tasks:

1. Click the **Tasks** tab.
2. Click **Add Task**.
3. Select **Normal Task**.
4. Click **OK**.





Normal Task *is used for specific cost items (such as motor, cylinder, service, inspection).*

Unassigned Task *is used to set up provisions for costs within a specific subsystem.*

Other Operating Cost Task *is used to set up other operating costs.*

5. Complete the mandatory fields on the **Base Details** tab.
6. On the **Job Details** tab, complete the **Job Header Details** on the **Job Details** tab (**Job Code**, **Cost Source**, **Parts**, **Labour** and **Misc. Cost** amounts, and the **Labour** and **Duration** fields).
7. A strategy task can have multiple jobs linked to it. The **Cost Source** options for a single job are:
   - **Manual**: You can manually enter the costs.
   - **Standard Job**: The costs are sourced from standard jobs in AMT. Refer to the Standard Job training document for more information.

- If the estimate has tasks that are linked to standard jobs, the estimate reflects the price of the standard job when it was first linked to the estimate.
  The prices are not changed because the estimate is designed to reflect a fixed cost per hour at a point in time, in order for consistent figures to be used during contract negotiations.
- Updates to the pricing of the standard jobs will not affect the estimate unless you manually refresh the estimate's standard job's costs.

- **Cost Allocation**: You can define multiple cost elements across different cost centres, expense elements, labour activities and work groups.
- **Consumables**: You can enter quantity, top up and unit price to calculate the parts cost element of the job.



There is no link between labour hours or labour activity and the calculation of labour cost.

The pricing date reflects the date at which the costs entered are valid. It is from this point that escalation will be applied.



You can apply pricing increases and decreases and other changes to tasks using various AMT multi-editors.

8. Complete the scheduling fields on the **Scheduling** tab (**Scheduling Method**, **UOM**, **First Occurrence**, **Frequency**, and **Not After** if required).



Refer to the Scheduling Methods training manual to learn more about AMT's Advanced Scheduling Logic options.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

5. Add a Strategy Task (if required)

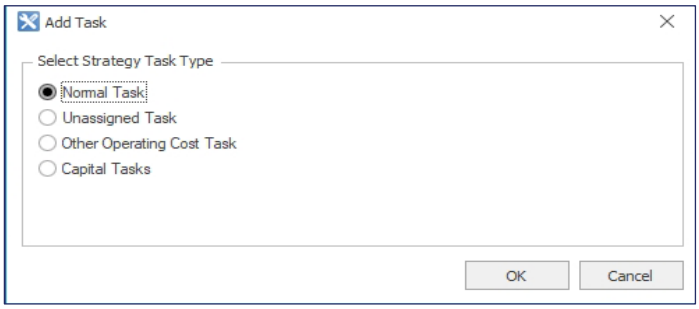
Overview

You may want to add an additional strategy task to the equipment, or if created from blank, you may want to add strategy tasks to the equipment.

Procedure – Add a Strategy Task (if required)

Perform the following steps to add a strategy task:

1. Click the **Tasks** tab on the **Edit Equipment** dialog. Then, click **Add** to create a new task.
2. The **Add Strategy Task** dialog displays. Select either a **Normal Task** or an **Unassigned Task**. In this example, select **Normal Task**.



3. The **Add Strategy Task** dialog displays. To define the strategy task, perform these actions on the **Base Details** tab:
    - Enter a **Description** for the task.
    - Select **Component Code**.
    - Select **Task Type**.
    - In the **Task Mode** section, select **Non Planning**.
4. Click the **Scheduling** tab and perform the following actions:
    - Select **Scheduling Method**.
    - Enter Task **First Occurrence:** The first occurrence value.
    - Enter Task **Frequency**
5. Click the **Job Details** tab and perform the following actions:
    - Select **Cost Source.** In this example, use Manual.
    - Enter **Labour Hours**.
    - Enter the amount of **Labour Cost**.
    - Select **Labour Expense Elements.**
    - Click **Save**.

    Notes:

    A strategy task can have multiple jobs linked to it. The **Cost Source** options for a single job are:

    - **Manual**: You can manually enter the costs.
    - **Standard Job**: The costs are sourced from standard jobs in AMT. Refer to the Standard Job training document for more information.
        - If the estimate has tasks that are linked to standard jobs, the estimate reflects the price of the standard job as when it was first linked to the estimate.

The prices are not changed because the estimate is designed to reflect a fixed cost per hour at a point in time, in order for consistent figures to be used during contract negotiations.

- Updates to the pricing of the standard jobs will not affect the estimate unless you manually refresh the estimate's standard job's costs.

- **Cost Allocation**: You can define multiple cost elements across different cost centres, expense elements, labour activities and work groups.
- **Consumables**: You can enter quantity, top up and unit price to calculate the parts cost element of the job.



There is no link between labour hours or labour activity and the calculation of labour cost.

The pricing date reflects the date at which the costs entered are valid. It is from this point that escalation will be applied.

Duration is the amount of time required to complete the job.



You can apply pricing increases and decreases and other changes to tasks using various AMT multi-editors.

- If you are not analysing the modelling equipment by work centre expense elements, cost centres and other codes, it is not necessary to set up the work centre.



*You can select the modelling fleet and the projection, and then apply the required labour rates across many tasks. To do this, from the **My Functions** hierarchy select **Modelling** > **Administration** > **Projections** > **Multi-Task Labour Editor**.*

6. Click the **Tasks** tab of the **Edit Equipment** dialog, and then click **Apply Filter** to refilter and view the new task.

   A recalculation notification dialog displays. Click **Yes** or **No** depending on your requirements.



7. AMT displays the updated list. Scroll down the task list to view the new task.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

6. Create a Standard Job (if required)

Overview

You can create a standard job to add more detail to the modelling equipment, such as the materials and labour required.

Procedure – Create a Standard Job (if required)

Perform the following steps to create a standard job:

1. Search **Standard Job Manager** using the **Search** field. Double-click **Standard Job Manager**.



*Alternatively, from the **My Functions** hierarchy, select **Knowledge Database** > **Standard Job Manager**.*

2. On the **Standard Job Manager (Priced Job)** dialog, populate the **Branch** field to find the modelling equipment.
    - Click **Apply Filter** to view the existing standard jobs for the selected branch.
3. The list of existing standard jobs displays. If required, you can create a new standard job by clicking **Add**.

4. The **Standard Job (Priced Job)** dialog displays:
    - **Description:** Enter the description for the standard job.
    - **Pricing Branch:** Modelling equipment is managed from the **Pricing Branch**. Select the branch under which the fleet of modelling equipment is linked. This determines which labour rates are relevant for the labour activities that you select in the standard job.
    - **Pricing Group:** Select the pricing group of the fleet to which the modelling equipment is linked.



- *AMT will be able to apply labour rates for the labour activity if it exists in the system table.*
- *Make note of the description of the standard job so that you are able to link it to the modelling equipment.*

5. Enter the required information in the **Standard Job Codes** section. The standard codes to select are:

    - **Model**: GENERAL PROCESS (unless you want to make it model-specific)
    - **Component Code**: NONE (unless you want to make it component code-specific)
    - **Modifier Code**: 00-(None) (unless you wish to make it modifier code-specific)
    - **Task Type**: Select one.
    - **Task Counter**: 0-(None)
    - **Job Code**: 000-None

6. Click the **Operation** tab, then click **Add** to add an operation to the job.

7. Click the **Parts** tab and then click **Drag & Drop Parts**.

8. The **Add Parts (Drag & Drop)** dialog displays:
   - Search for the required part using the **Part Number** field or the **Part Description** field. Both are wildcard searches. These actions search for matches in the AMT Parts Manager table.
   - Click **Apply Filter**.
   - The search results display.

9. Drag and drop the required part to the **Standard Job (Priced Job)** dialog.

10. The **Standard Job (Priced Job)** confirmation dialog displays. Click **Yes**.

11. The **Parts** table is populated. Enter the quantity required and the probability that the part will be used in the **Qty** and **Probability %** columns, respectively.

    - Check the **Unit Price** and the **Total Cost** of the part.
    - You can also select the relevant expense elements in the **Expense Element** column.



The probability that the part will be used is how likely the part is going to be required, for example 50% can be interpreted to mean we only need this part once every two times this standard job is performed. The impact from a costing point of view is to reduce the parts costs of the standard job and thus the strategy task to which the standard job is linked, and thus the budget.

12. Click the **Labour** tab and then click **Add** to add labour.
    - If **Labour Activity** is selected, AMT automatically applies the labour rate, that is, the value in the **Base Rate/Hrs** column.
    - If **Labour Activity** is not selected, you can enter a description in the **Description** column and manually enter values in the **Base Rate/Hrs** and **Hrs** columns.
    - You can also select the relevant expense elements in the **Expense Element** column.



- *In this step, the labour rate is being estimated for future equipment. Modelling equipment provides this flexibility.*

13. Click **Save and Close**.



After saving, AMT creates one standard job, which is priced for the combination of Branch-Pricing Group-Currency. This is called the "Priced Job".

AMT prices a standard job for the various Branch-Pricing Group-Currency combinations which is initially created and then used, that is, linked to a strategy task(s). The priced jobs are what you see in Standard Job Manager, so you can see not only the components of the standard job but also its pricing that is, how much it costs.

14. In the **Edit Equipment** dialog, double-click the task to which the Standard Job is to be linked.

15. The **Edit Strategy Task** dialog displays. Click the **Job Details** tab.

16. In the **Job Header Details** section of the **Job Details** tab, set **Cost Source** as **Standard Job**. The **Standard Job** field displays.

17. Click the **Search** icon against the **Standard Job** field.

18. The **Select Standard Job** dialog displays. In the **Filters** section, enter the relevant details to locate the desired standard job. Click **Apply Filters**.

Notes:

- AMT defaults to searching jobs that have the same branch, price group and currency as that of the strategy task. Remember, the strategy task (which has a currency) is linked to an equipment, which is linked to a fleet (which references the pricing group for parts), which in turn is linked to a site, and then linked to a branch (which holds the labour rates).
- It is possible to select a job from a different price group. If you do this, AMT will create a new priced job from the existing standard job, which will be linked to the price group and priced for that price group and branch.
    - The underlying detail of the standard job does not change, such as the list of parts, quantities, labour activity, labour hours and so on. Only the cost of the standard job changes because of the Branch-Pricing Group-Currency combination of the strategy task.

19. AMT populates the corresponding list. Double-click the required **Standard Job**.

    The **Standard Job** field of the **Edit Strategy Task** dialog is now populated. The **Job Costs** section displays the information in the selected standard job.

20. Click **Save**.

21. Under the **Tasks** tab of the **Edit Equipment** dialog, click **Apply Filter** to refilter and recalculate the task list.

    - The **Cost Source** of the task is a standard job created in AMT. In this example, the standard job is "SJ-AMT". If the standard job was created in the ERP, the **Cost Source** column displays "SJ-EPF", where EPF is the integration layout.
    - The summary calculations at the bottom right side of the dialog reflect the updated details.

    The costs for the newly created task are forecasted. The costs for the updated standard job are also updated.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

7. Create Alternate projections for Modelling Equipment (if required)

Procedure – Create Alternate projections for Modelling Equipment (if required)

Perform the following steps to create alternate projections for modelling equipment:

1. On the **Edit Equipment** dialog, click **Refresh** to refresh the values in the **Equipment Projection List** section.

2. Click **Add** in the **Equipment Projection List** section to create a version of the projection. Ensure that the **Multi-Projection** check box in the **Enabled** section is selected.

3. On the **Add / Copy Projection** dialog, perform the following actions:
   - **Select a Projection Type to Create** section: Set the **Projection Type** as **Estimate-Draft**.
   - **Select Projection Source** section: Select the **Use Existing (Select Source of Projection)** option and enter the required information.
   - **Select Projection Destination** section: Rename the projection and update the pricing date in the **Projection name** and **Pricing Date** fields, respectively.
   - Click **Process**.



The pricing date for the projection must be less than or equal to the current date.

4. The **Copy Projection** dialog displays. Select the **Leave the pricing as is** option and click **OK**.

5. The **Add / Copy Projection** confirmation dialog displays. Click **OK**.

6. In the **Equipment Projection List** section of the **Edit Equipment** dialog, click **Refresh**.

7. Click the **Tasks** tab.

8. Enter the name of the copied projection in the **Projection** field and click **Apply Filter** to refilter and recalculate the costs.

9. Return to the **Base Details** tab. You will see that the values corresponding to the copied projection are updated.

10. Double-click the current **Projection Name** to make additional changes to the selected projection.

11. The **Edit Projection** dialog displays. Make the relevant changes to the projection and click **Save and Close**.

8. Adjust Strategy Tasks

Procedure – Adjust Strategy Tasks

A number of administration tools exist in AMT to help you add, modify, copy and delete projected tasks more easily. Consequently, managing estimates is efficient and user-friendly.

The tools include:

- **Multi-Task Cost Adjuster**: Adjust costs (and duration and labour hours) of groups of tasks (for example, increase all parts prices for PM tasks for the fleet of 5 equipment by 3%). This is limited to normal and unassigned tasks.
- **Multi-task Scope Adjustor**: Adjust the cost of strategy tasks linked to standard jobs by applying factors. You can also adjust travel (labour and miscellaneous) costs. This is limited to normal and unassigned tasks.
- **Multi-Task Editor**: Edit any details of a task for multiple equipment (such as change the task type for the radiator strategy task for all 8 trucks in the fleet). This is limited to normal and unassigned tasks.
- **Multi-Equipment Delete Task**: Delete a task from many equipment projections.
- **Add / Copy / Delete Projection**: Copy a projection from one equipment to one or more equipment. This is limited to normal and unassigned tasks.
- **Copy Projected Task**: Copy a single strategy task from one projection to many projections (if the task is linked to a standard jobs it will retain the link).

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

9. Define Usage Assumptions

Overview

You can link the **Annual Utilisation** to a usage profile to represent the future mine plan for when the assets come online.

Procedure – Define Usage Assumptions

Perform the following steps to define usage assumptions:

1. Search **Usage Profile Manager** using the **Search** field. Double-click **Usage Profile Manager**.



*Alternatively, from the **My Functions** hierarchy, select **Modelling** > **Administration** > **Usage Profile > Usage Profile Manager**.*

The **Usage Profile Manager** dialog displays.

2. Click **Add** to add a new usage profile.



In modelling, a usage profile can be applied to more than one equipment.

3. The **Add Usage Profile** dialog displays. Enter a unique name for the usage profile in the **Usage Profile** field.
    - In the **Unit of Measure (UOMs)** section, click **Add** to add a unit of measure.
4. The **Unit of measure** dialog displays.
    - Enter a valid value in the **Usage per parent UOM** field.
    - Select the **Unit of Measure.**
    - Click **Save**.
5. The entered information displays in the **Unit of Measure (UOMs)** section and is added as a column in the **Usage Steps** section.
    - Click **Add** in the **Usage Steps** section if you require steps in the usage profile.
6. On the **Add Usage Step** dialog, select the required date in the **Step Start Date** field to model the profile for the mining plan. Add more steps to the profile clicking **Add** as required.
7. Click **Save**.

You can edit the details for the unit of measure according to the mining plan. You can also add more steps to the profile by clicking **Add**.

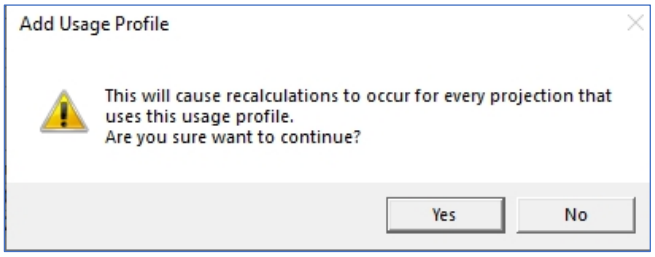
8. You can add more units of measure in modelling in addition to the hours that drive the scheduling of tasks.

    - Click **Add** in the **Unit of Measure (UOMs)** section to add more units of measure.
    - Edit the required entries for the new units of measure in the **Usage Steps** section.
    - Click **Save**.

*New units of measure can be added to AMT using the **Unit of Measure** system table.*

*UOMs can be a child of **Calendar Year** or can be a child of another UOM. For example, Fuel Burn could be child of Machine Hours so that for every 1 machine hour "x" amount of fuel is consumed.*

9. The **Add Usage Steps** confirmation dialog displays. Click **Yes**.



This usage profile has not been linked to any equipment yet.

10. To link the created usage profile to an asset, in the **Edit Equipment** dialog, select the **Advanced utilisation** option in the **System Configuration** section.

11. Click the **Term** tab. In the **Usage Profile** field, select the created **Usage Profile**. Click **Refresh**.

12. The **Edit Equipment** confirmation dialog displays. At this point, the projections will be linked to the created usage profile. Click **Yes** to confirm.

This dialog will not appear when strategy tasks don't need to be recalculated. However, the updates you made will still be saved.

Usage profiles can be updated at any time. Changes to usage profiles will impact the end usage and projections of the equipment to which the usage profile is linked.

13. In the **Term** tab, the information from the usage profile has added the hours, cycles, and tonnes to the **Start Usage** field instead of the **Annual Utilisation** field. The value in the **End Usage** field remains the same, but the end date has changed.

14. Click the **Tasks** tab. Click **Apply Filter** to recalculate the values based on the usage profile.

- *In the summary calculation on the bottom-right corner, the usage hours are the same, but the timing of the tasks will change because of the usage profile.*
- *Over different periods of time, utilisation may decrease or increase. Therefore, tasks driven by utilisation will happen either sooner or later than the original assumption.*

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

10. Set Availability Forecast for Modelling Equipment

Overview

To perform availability forecasting, task durations need to be defined and the fleet should be updated.

Procedure – Availability Forecast for Modelling Equipment

Perform the following steps for availability forecasting for modelling equipment.

1. Search **Fleet Administration** using the **Search** field. Double-click **Fleet Administration**.



*Alternatively, from the **My Functions** hierarchy, select **Modelling** > **Administration** > **Equipment Hierarchy** > **Fleet Administration**.*

2. On the **Fleet Administration** dialog, click **Apply Filter** to see the search results and double-click the relevant item.

3. The **Edit Fleet** dialog displays. Ensure that the formula in the **Availability Formula** section is set to **Availability Type** = **Physical Availability**.

4. Click **Availability Calculation**.

5. On the **Availability Calculation Parameters** dialog, perform the following actions:

   - Select the **Enable Forecasting Availability** check box.
   - Enter the number of **Annual Scheduled Hours:** the assumption of how many hours each equipment in the fleet will operate per annum.
   - Enter the **Scheduling Efficiency**: the assumption of how efficient maintenance planning will be performed. That is, will the planner group all major tasks, which are forecasted to occur in the same month, to be executed at the same time (the absolute minimum stoppages ~ 100%) or will each task be performed at an independent stoppage (maximum possible stoppages ~ 0%). Usually this is set to 50%.
   - On the **Task Specific Downtime** section, select the check boxes corresponding to the tasks that will draw down on full efficiency. This may not cover all potential downtimes. For example, if 95% is set as the Annual Base Downtime rate, and you select component change outs and PM services to be included (because they will be performed under a contract), the availability rate will drop because the downtime for change outs and PM services will be factored into the availability forecast.
   - On the **Annual Base Downtime (Availability)** section, enter the task downtime for each year depending on the age of the machine. This is the base downtime that accounts for small and regular maintenance. Usually the rate is high (> 95%) if only minor breakdown-type repairs were to be performed.

Notes:

- AMT will forecast the availability based on the base downtime and task-specific downtime.
- Scheduling efficiency considers how efficiently maintenance planning will be performed:

    - will the planners group tasks, which are forecast to occur in the same month, to be executed at the same time (the absolute minimum stoppages = Scheduling Efficiency 100%), or
    - will each task be performed at an independent stoppage (maximum possible stoppages = Scheduling Efficiency 0%).
- For example, Task1 Duration = 100 hours and Task2 Duration = 8 hours; both occurring in the same month, therefore Scheduling Efficiency:
    - 0% = 108 hours (100+8)
    - 100% = 100 hours (that is, tasks would be aligned under the task with the greatest duration)
    - 50% = 104 hours of downtime (100 + 50% of 8).

6. Click **Save**.

11. Rollover of Real Equipment to Modelling Asset

Overview

To budget the movement of an equipment from site a to site b, and, while doing so, change details of the equipment/strategy tasks/usages before adding into budget b, we recommend that you use the Rollover process.

This rollover process allows you to create a modelling equipment, recognising when the strategy tasks are next due.

Procedure - Rollover of Real Equipment to Modelling Asset

Perform the following steps to roll over the modelling equipment:

1. Click **Modelling Equipment** on the **Equipment Navigator**.
2. Click **Apply Filter**.
3. On the **Equipment Selector** box, select the desired equipment in the **Equipment** field. Click **OK**.
4. The selected equipment is listed in the **Equipment Navigator**. Double-click the new equipment to view its details.



If this is a usage-based equipment, the start usage of the modelling equipment must equal the start usage of the source real equipment. This means the units of measure set up for the modelling equipment must match that of the source real equipment.

5. On the **Edit Equipment** dialog, click the **Base** Details tab, and click **Add** in the **Equipment Projection List** section.
6. The **Add / Copy Projection** dialog displays.
    1. Select **Projection Type**.
    2. In the **Select Projection Source** section:
        - Select the **Eqp Type**.
        - Select the **Equipment**.
        - Select the **Projection**.
    3. In the **Select Projection Destination** section, enter an appropriate name in the **Projection name** field.
    Whenever you create a new projection for a modelling equipment, any previous projections are saved as drafts before being overwritten by the new projection. You can update the name of the existing projection. **The existing Estimate-Current projection will be renamed**.
    4. Select the **Apply Advanced Rollover Equipment Logic** check box.
7. The **Advanced Rollover Details** section displays.
    - Select the equipment in the **Select Equipment as a basis for the advanced rollover calculations** field.

- Select the **Set First based on Rollover Equipment Last Occurrence** option.
- Click **Process**.

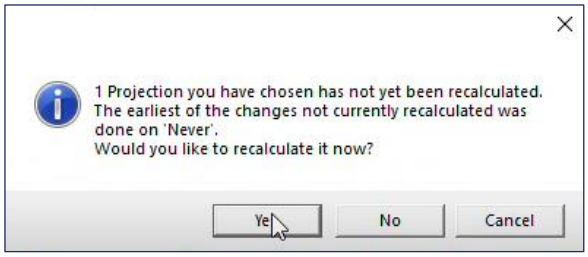
8. The **Copy Projection** dialog displays. If there are any discrepancies in the data entered, the dialog alerts you. You can choose to escalate or de-escalate the discrepancy, or take no action.

    - Select the **Leave the pricing as is with the new pricing date of** option.
    - Click **OK**.

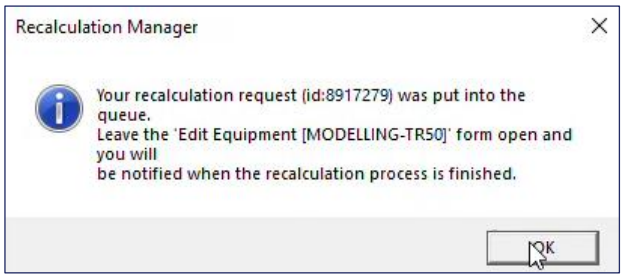
9. The **Add / Copy Projection** dialog displays confirming that the process is successfully complete. Click **OK**.

10. Click the **Tasks** tab and then click **Apply Filter**.

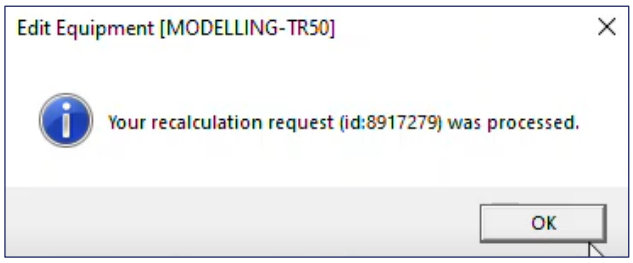
11. A recalculation notification box displays asking whether you would like to recalculate the projections. Click **Yes**.



12. The **Recalculation Manager** dialog displays. Click **OK**.



13. Once the recalculation is complete, the **Edit Equipment** notification box displays. Click **OK**.



14. AMT populates the recalculated strategy tasks list.



The rollover process recognises that for an existing equipment some work is already complete and forecasts when the next strategy task must be performed, for example, a component was replaced and is due for replacement every year. The rollover adjusts the first column to reflect when the strategy task is next forecast to be performed.

15. If a component frequency needs to be updated as per the requirements of the new site (for example, site B), double-click the **Frequency** corresponding to that strategy task.

16. The **Edit Strategy Task** dialog displays.

    - Click the **Scheduling** tab.
    - Enter the appropriate value in the **Frequency** field.
    - Click **Save**.

Notes:

- The strategy tasks, and any linked standard jobs, are copied into the modelling equipment.
- When a strategy task, linked to a standard job, comes into a modelling equipment from another equipment, such as a centreline, the cost values on the modelling equipment's strategy task are a snapshot of the cost values of the source equipment's strategy task.
    - The cost values of modelling equipment are not automatically updated to reflect the modelling equipment's labour rates sourced from the branch or parts pricing group sourced from the fleet. Unless the modelling equipment is under the same Branch-Fleet as the source equipment.
    - Moreover, the costs values are not automatically updated even if the linked standard job is updated. This is a design principle of modelling equipment in AMT.
- If you want to see the strategy task updated with the latest standard job cost, then follow the steps below:

    - Search **Refresh Standard Job Pricing** using the **Search** field and double-click **Refresh Standard Job Pricing**. Alternatively, from the **My Functions** hierarchy, select **Modelling > Administration > Projections > Refresh Standard Job Pricing**.
    - The **Refresh Standard Job Pricing** dialog displays. Select the appropriate options from the **Fleet**, **Equipment** and **Projection**.
    - Populate the **Task Type** and **Strategy Tasks** fields.
    - Click **Refresh Pricing**.

17. In the **Edit Equipment** dialog, click **Save**.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

12. Change Modelling Equipment's Cost Centre

Overview

It may be necessary to change the cost centre of the equipment so that it aligns to the site where the equipment will move in the future.

Procedure – Change Modelling Equipment's Cost Centre

Perform the following steps to change modelling equipment's cost centre:

1. Open the modelling equipment and click the **Classifications** tab.
2. Select the appropriate **Cost Centre** and click **Save**.



*All strategy tasks of the modelling equipment will inherit the **Equipment Cost Centre** unless a strategy task has been assigned a cost centre directly, in which case the strategy task's cost centre will also need to be updated.*

13. Change Cost Centre for Individual Tasks

Overview

Perform the following steps to change the cost centre linkage for individual tasks (if required):

1. In the **Tasks** tab of the **Edit Equipment** dialog, double-click the applicable strategy task in the **Description** column.
2. On the corresponding **Edit Strategy Task** dialog, click the **Job Details** tab.
3. From the **Cost Centre** field, first remove the linkage to the existing cost centre by clicking the **Remove** (**X**) icon and then link the task to another cost centre or leave it blank.
4. Click **Save**.



If the Cost Centre field is left blank, AMT will assume the task is to be linked to the default cost centre of the equipment.

14. Change Cost Centre for All Tasks

Procedure – Change Cost Centre for All Tasks

1. On the AMT home page, search **Multi-Task Editor** using the **Search** field. Double-click **Multi-Task Editor**.



*Alternatively, from the **My Functions** hierarchy, select **Modelling** > **Administration** > **Projections** > **Multi-Task Editor**.*

2. On the **Multi Strategy Task Editor** dialog, perform the following actions:

   - **Eqp Type** field: Select **Modelling**.
   - **Equipment** field: Select the relevant Modelling **Equipment**.
   - **Projection** field: Select the **Projection** to which you want to apply the changes.
   - **Job Filters** section: Select the **Edit Job Details** check box.
   - **Display Fields** field: Select the **Cost Centre** option.
   - Click **Apply Filter**.

3. AMT populates the corresponding list of items.

   - To search and assign a common cost centre to all the tasks, in the **Strategy Task Modification** section, select the desired cost centre in the **Cost Centre** column.

4. Click **Apply Changes** to execute the changes across all tasks.

   A confirmation message displays.

5. Click **OK**.

6. Click **Apply Filters** to see the changes in the results grid.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

15. Validate Life Cycle Costs

Procedure – Validate Life Cycle Costs

Use the reporting functionality to validate the modelling estimate. Use the **Projection Summary** and **Projection Details** reports as a starting point. Other reports recommended for analysis using AMT:

- Total cost
- Productivity
- Cost per UOM (Cost per Hour / tonnes)
- Major cost items
- % Unscheduled costs
- Labour ratio
- Availability
- Cost breakdown (by system, task type, expense element)

In addition, the costs of the modelling projection should be validated against the source data. The following reports will help this process:

- Projection Summary - Modelling
- Projection Details - Modelling
- Cost Analysis “Life Cycle” – Modelling
- Cost per UOM per Interval – Modelling
- Cost per Interval – Modelling (example below)
- Cost per UOM per Interval – Modelling.

After the baseline is established, the manager may want to create a more detailed, customised modelling asset, perhaps to be specific to a particular application or refine the way in which components’ lives are measured or forecast.

Equipment and estimates can be benchmarked against each other in many reports by selecting **Equipment** as the **Analyse by** in the report parameters.

16. Link Component Structures

Overview

If the CBM (Condition Based Maintenance) module is installed, you may want to create component structures to further refine strategy task cost projections. A component structure defines the failure modes for a specific component (strategy task) for a specific model of equipment. Component structures can only be created if there is sufficient knowledge of the component's failure modes.

Linking a strategy task to a component structure does not override the task's existing scheduling parameters, nor its costs. Rather, the component structure is recalculated using the task's original values using ratios.

Ultimately, linking to a component structure results in:

- A more detailed definition of the maintenance strategy for this task, which can be viewed by clicking **Quick Links**.
- The ability to create a repair reserve.
- Defined CBM measurements for each failure mode, which will be useful when the centreline is applied to a real equipment/asset.

Refer to the CBM training document for the steps to create and link component structures to strategy tasks.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

17. Create an Overhead Modelling Projection

Overview

Overheads costs can be added as part of the total maintenance cost calculation. This method is captured in the estimate-current projection's cost schedule (which has been discussed previously).

Before setting up overheads, it is important to validate:

- How are overheads costed into the site, project, or contract - part of the direct maintenance costs, or separate?
- If AMT is being used to track and forecast maintenance overhead costs for real equipment, how should overhead cost estimated be defined for modelling equipment if at all?

The questions above are important considerations before you start setting up overheads in AMT.

The first step is to ensure the appropriate overhead costs categories have are set up in AMT's system table; use the function **Overhead Cost Category**.

Procedure – Create an Overhead Modelling Projection

Use AMT's **Overhead Navigator** function to create an overhead projection. Perform the following steps to create an **Overhead Modelling Projection**:

1. On the AMT home page, search **Overhead Navigator** using the **Search** field. Double-click **Multi-Task Editor**.



*Alternatively, from the **My Functions** hierarchy, select **Modelling** > **Administration** > **Overheads** > **Overhead Navigator**.*

2. On the **Overhead Navigator**, click **Add**.

   On the **Overhead Header Editor screen**, perform the following actions:

3. Select the appropriate **Overhead level**.
   - **Overhead Level**: depends on the level that overheads are costed For example, do the overhead costs apply to specific fleets or a specific branch or site?
4. Select a **Level description**.
   - The options available depend on the overhead Level chosen (for example, if the **Overhead Level** is **Site**, a list of sites that exist in AMT are listed.
   - Set the **Start Month** and **End Month**.
   - Select **Projection** to **Estimate-Current**.
   - Select a **Currency**.
   - Only check **Include MARC billings** if the estimate is for a maintenance contract.
   - Click **Save**.
3. Populate the overhead's schedule with costs. This can be done by entering values in the grid or by copying and pasting values from Excel.



*A Billings revenue line will not appear if **Include MARC Billings** was not set.*



To copy and paste from Excel, the overhead projection has to be created in AMT in terms of the correct number of rows (representing cost categories) and columns (representing months).

18. Summary

Summary

The key points to remember from this guide are:

- Modelling equipment can be created using strategy tasks and standard job references from centrelines or real equipment. The life cycle term of a modelling equipment is defined within the equipment itself. It is not copied from another equipment.
- Modelling equipment can be imported into a budget and/or a forecast to factor in maintenance and labour costs of potential new assets and decide whether the new assets will meet the business need.
- You can perform the following create modelling assets steps in AMT:
    - Create Modelling Equipment
    - Define life-cycle term and confirm appropriate start date and usage:
        - Adding a Strategy Task, if required
        - Creating a Standard Job, if required
        - Creating Alternate projections for Modelling Equipment, if required.
        - Availability Forecasting for Modelling Equipment
    - Define usage assumptions.
    - Rollover of Real Equipment to Modelling asset
    - Edit the Standard Job
    - Unlink a Strategy Task from a Standard Job
    - Change Modelling Equipment's cost centre.
    - Change cost centre for individual tasks.
    - Change cost centre for all tasks.